Week 1 Assignment

Language of Computation

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My main takeaway from the article (and this assignment) is just how critical data management is and can be. Shocked is probably the best word to accurately describe the findings of the article. I don't know if it was simply a rush to publish, a clerical oversight, or negligence on part of the authors of the research, but the prevalence of data errors across so many published articles highlights just how critical and pervasive this issue is. For published, scientific journals, finding gene name errors in nearly 20% of the sample size tells me that there is something fundamentally flawed with the system. That quantity of errors/mistakes would be grossly unacceptable in just about every other industry.

This week's assignment made it very clear as to just how easily such errors can occur though. (Falsely) assuming that the data perfectly translates from one file type to another is something that I am now very much aware of. In the very small sample of the data we input into our files, seeing just how much the zip codes of the original .csv file I created and the dates of the bank transaction data was genuinely eye opening to me. It became very apparent just how quickly the source data can be compromised/inaccurate if careful attention is not paid to the transition between file types.

My personal experience with these issues is very limited. I do not normally work with large data sets. I don't really manage very many spreadsheets or anything of that nature. My experience is limited to either trying to open up an excel file in google sheets, or open a google sheet in excel. I've experienced a few minor frustrations when doing this, but these have mostly just been formatting issues.

After reading this material and doing this assignment, I do have a very practical application though. I perform the accounting for my business by utilizing quickbooks. For the most part, quickbooks links to my various bank and credit card accounts. However, there are at least two accounts where I have to go in and download my monthly statements so that I can then upload those files into quickbooks. These files are .csv files, and until last week's class I did not really know what that file type was. With this new understanding and knowledge, I am definitely going to be double checking the data moving forward though. Even minute changes in my accounting data would impact my tax liability.

Perhaps on a more fundamental level though, this assignment has just highlighted to me the importance of the data. First and foremost, the data has to be good. However, just having good data is not enough. The good data must be accurately ingested in order for the output to have any meaning or validity. This is a very important concept to remember as we move forward in this program.